

Lecture 5: The Rights and Responsibilities of Engineers

HUM 4441: Engineering Ethics

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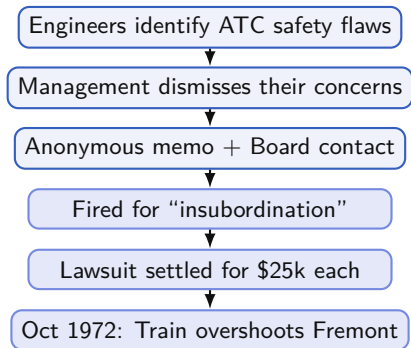
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The BART Case: Rights vs. Responsibilities in Conflict

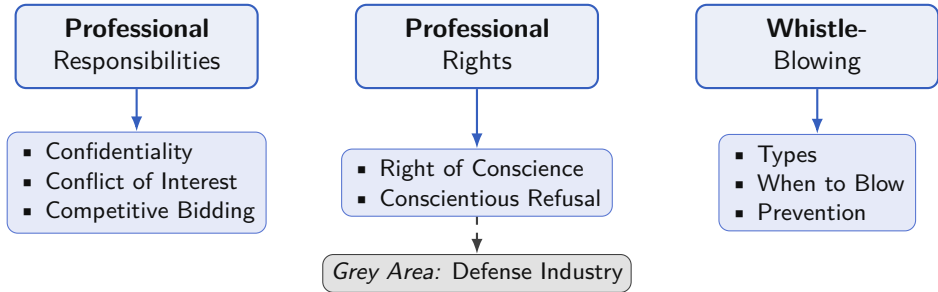


In the early 1970s, three BART engineers - Hjortsvang, Bruder, and Blankenzee - raised concerns about the automated train-control system's untested components.

Their efforts to protect the public cost them their jobs. The IEEE later filed an *amicus curiae* brief arguing the engineers were fulfilling their **professional duty**.

An engineer has a **duty** to protect the public *and* a **right** to have concerns taken seriously; without risking employment. Fleddermann, Ch. 6

Chapter Overview



Professional Responsibilities

1 Confidentiality

Engineers must keep employer/client information secret:

- Test results & designs
- Supplier identities
- Marketing strategies
- Production costs

Obligation continues *after* leaving
- courts balance this against career mobility.

2 Conflict of Interest

A situation where an engineer's personal, financial, or other interests could interfere with their professional judgment or duty to employer/client. Three types:

- **Actual** - compromises judgment now
- **Potential** - could become actual
- **Apparent** - appears conflicted

Resolution: disclose, recuse, or seek guidance.

3 Competitive Bidding

The practice where engineers compete for projects by submitting price proposals, rather than being selected solely on qualifications. Legalised by **U.S. Supreme Court (1978)**.

Ethical pitfalls:

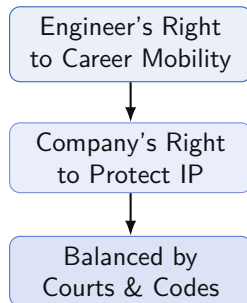
- Lowballing bids
- Overstating qualifications
- Disparaging competitors

Confidentiality: Where the Gray Areas Lie

The Post-Employment Problem

When an engineer changes jobs - especially to a competitor - she carries *experiential knowledge*: what works, what materials to choose, what to avoid.

- Legally, confidentiality survives employment
- Practically, knowledge cannot be “unlearned”
- Courts try to **balance** individual career rights against company IP rights
- The burden lies with **the individual engineer**



General Rights

Independent of professional status:

- Right to privacy
- Right to participate in activities outside work
- Right to object to company policies without fear of retribution
- Right to due process

Two Fundamental Professional Rights

1. Right of Professional Conscience

The right to exercise professional judgment in an ethical manner.

2. Right of Conscientious Refusal

No employer may pressure an engineer into behaviour she considers unethical.

Clear cases: Falsifying test results, misrepresenting safety

Less clear: Refusing defense or environmentally hazardous work

Employers should **reasonably accommodate** such requests.

A Legitimate Choice Either Way

- Defense is one of the largest global employers of engineers
- **Choosing not to work** in defense - ethically justifiable: remoteness from killing does not remove moral responsibility
- **Choosing to work** in defense - also justifiable: national defense is a legitimate governmental function

Guidance

Avoid any project you deem unethical - even if it means forgoing career advancement.

Questions to ask before accepting defense work:

Is this consistent with my values?

Will it be sold only to allies?

Does it serve a defensive purpose?

Am I prepared to see this through?

Fleddermann,

Ch. 6, §6.3.1

Whistle-Blowing

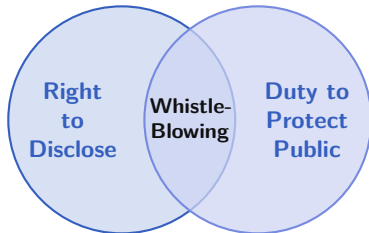
Definition

Whistle-blowing is the act by an employee of informing the public or higher management of unethical or illegal behavior by an employer or supervisor.

It straddles the line between **right** and **responsibility**:

- Engineers have a *duty* to protect public health and safety
- Engineers have a *right* to disclose wrongdoing and expect action

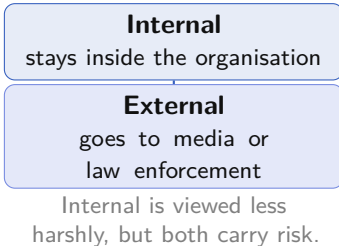
Perceived by organisations as extreme **disloyalty** - like a teammate calling a penalty on their own team.



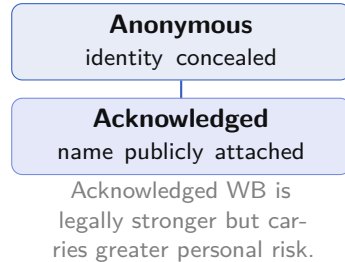
Fleddermann, Ch. 6, §6.4

Types of Whistle-Blowing

By destination



By disclosure



BART: Engineers sent an **anonymous** internal memo - which escalated to **external** WB when a board member leaked to the press.

When to Whistle Blow?

All four conditions must be met before whistle-blowing is morally permissible.

1 Need

A **clear and important harm** that can be avoided. Proportion matters - patterns of small wrongs can add up.

2 Proximity

Must have **firsthand knowledge** and sufficient expertise. Hearsay is not adequate.

3 Capability

A **reasonable chance of success**. Not obligated to risk your career if you cannot see the case through.

4 Last Resort

All **internal channels exhausted** and no one more capable can act. WB is the final option.

Obligation vs. Permission

You *may* blow the whistle when all four conditions are met. You are *obligated* only when **imminent and serious harm** is likely and all four conditions hold.

Harris, Pritchard & Rabins, 2000

Acceptable and Unacceptable Motives

Acceptable motives

- Protecting the health and safety of the **public**
- Stopping ongoing harm that management ignores
- Fulfilling professional ethical duties under engineering codes

Unacceptable motives

- Exacting **revenge** on supervisors or colleagues
- Seeking future **financial gains** (book deals, speaking fees)
- Personal grudges unrelated to public welfare

Before blowing the whistle, a great deal of **introspection and reflection** is required.

Preventing Whistle-Blowing: An Employer's Perspective

Four organisational measures that reduce the need to go outside the organisation.

1

Ethics Culture

Top management must demonstrate commitment to ethics through training and ethical conduct.

2

Clear Communication

Establish open internal channels allowing employees to raise concerns without pressure.

3

No Retaliation

Employees must have access to senior management with a guarantee of no retribution.

4

Admit Mistakes

Management should acknowledge errors publicly, setting an ethical tone throughout.

What *not* to do

Firing whistle-blowers and intimidating potential ones is both **ineffective** and **ethically unacceptable**.

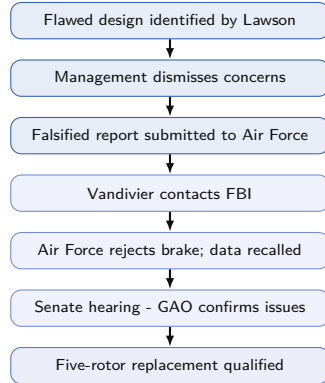
Case Study: The Goodrich A7-D Brake

Background

- B.F. Goodrich won a **1967 contract** for A7-D aircraft brakes
- Innovative **four-rotor** design - lighter than five-rotor alternatives
- Tight deadline: flight-ready by **June 1968**

What went wrong

- Lawson found no lining material could meet Air Force specs
- Management ordered a report based on **falsified test data**
- Technical writer Vandivier objected - and was overruled



Case Study: Hartford Civic Center Roof Collapse (1978)

What happened

On 18 January 1978 at 4 a.m., the arena roof collapsed under just 4.8 inches of snow - well within its design load. Fortunately the arena was empty; hours earlier it held thousands of spectators.

Contributing factors

- Top chord members overloaded by up to **852%**
- Frame weight **23 lbs/sqft** vs. designed 18 lbs/sqft
- Excessive sagging repeatedly reported - and dismissed
- Mid-point bracing omitted; not caught in inspection
- Over-reliance on unvalidated **1970s computer software**

Ethical Failures

- Engineers **ignored repeated deflection warnings**
- “Discrepancies between actual and theoretical should be expected” - a rationalisation that put lives at risk
- Design review procedures failed to catch the original error

Lesson

Physical warning signs cannot be explained away. Field evidence must be taken as seriously as computer models.

Ross, 1984; Martin, 2001

Summary

Responsibilities

- Keep information **confidential**
- Manage **conflicts of interest**
- Compete **honestly** in bidding

Rights

- **Professional conscience**: ethical judgment
- **Conscientious refusal**: say no to unethical tasks
- Right to **object** without retribution

Whistle-Blowing

- Permissible: all four conditions hold
- **Obligatory** when imminent harm is at stake
- Motive: **public protection**, not revenge
- Prevented by ethics culture & open communication

Rights and responsibilities of engineers are deeply intertwined - and both must be respected.

Additional Case Studies

Reading

- Fleddermann, Chapter 6 - *The Rights and Responsibilities of Engineers* (complete)
- Focus on the **BART case** and **Goodrich A7-D** sections

Question

In the BART case, were the three engineers *morally obligated* to blow the whistle, or did they only have *permission* to do so? Apply the four conditions from §6.4.2 in your answer.

Case Studies

- **The BART Case** - internal vs. external WB; consequences for engineers
- **Engineers and the Defense Industry** - right of conscientious refusal
- **Optional:** Hartford Civic Center Collapse - responsibilities under design review

Thank You